SUPPLEMENTARY MATERIAL FOR META-ANIMATOR

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In this supplementary material, we first review the fundamental concepts underlying our framework in Section A and discuss related work in Section B. Section C provides implementation details, Section D describes the Meta-Tasks, and Section E outlines the data curation process. Section F compares the training resources required by different methods. Section G then presents additional experiments on Meta-Tasks during few-shot fine-tuning, text-guided property modification, blur operators, and reference-set size. Finally, we discuss efficiency and practical adaptability in Section H and present additional visualization results in Section I.

A. PRELIMINARIES

This section introduces the core components underlying our method, including latent diffusion models, DDIM sampling and inversion, ControlNet, and condition-specific LoRAs. These foundations enable efficient and controllable image generation in our framework.

Latent Diffusion Models. The latent diffusion models offer a novel approach by performing denoising in the latent space instead of the image pixel space [1, 2], mitigating computational costs without compromising image quality. Stable Diffusion (SD) model [3] is a canonical instance of the Latent Diffusion Model (LDM) paradigm, which consists of a Variational AutoEncoder (VAE) [4] and a time-step conditioned U-Net [5].

Formally, the VQ-VAE encoder maps an input image $\boldsymbol{x}$ into a latent space representation $\boldsymbol{z} = \mathcal{E}(\boldsymbol{x})$, while the decoder reconstructs an image $\boldsymbol{x}'$ from a generated latent code $\boldsymbol{z}'$ as $\boldsymbol{x}' = \mathcal{D}(\boldsymbol{z}')$. The denoising process involves T successive steps to transform the initial Gaussian noise $\boldsymbol{z_T} \in \mathcal{N}(0, I)$ into the desired latent space features $\boldsymbol{z_0}$, which can be projected onto the RGB image space via the VQ-VAE decoder. The objective of each denoising iteration is to estimate the noise residual between the current latent $\boldsymbol{z_t}$ and the target latent $\boldsymbol{z_0}$, which is commonly accomplished using U-Net, with the loss function of

$$\mathcal{L} = \mathbb{E}_{\mathcal{E}(\boldsymbol{x}),c,\epsilon\sim\mathcal{N}(0,I),t}\left[\|\epsilon - \epsilon_\theta\left(\boldsymbol{z_t}, t, c\right)\|_2^2\right], \tag{1}$$

where $\epsilon_\theta(\cdot)$ denotes the U-Net for noise prediction, c denotes the conditioning input, t the time-step, and $\boldsymbol{z_t}$ the intermediate result at time-step t in the denoising process.

DDIM Sampling and Inversion. During denoising steps, a deterministic DDIM sampling is always used to sample a denoised $\boldsymbol{z_0}$ from standard Gaussian noise $\boldsymbol{z_t}$. Formally, a latent code $\boldsymbol{z_{t-1}}$ is generated from a latent code $\boldsymbol{z_t}$:

$$\begin{aligned}\boldsymbol{z_{t-1}} &= \sqrt{\alpha_{t-1}}\underbrace{\left(\frac{\boldsymbol{z_t} - \sqrt{1-\alpha_t}\epsilon_\theta\left(\boldsymbol{z_t}, t\right)}{\sqrt{\alpha_t}}\right)}_{\text{predicted } \boldsymbol{z_0}} + \\ &\underbrace{\sqrt{1-\alpha_{t-1}-\sigma_t^2}\cdot\epsilon_\theta\left(\boldsymbol{z_t}, t\right)}_{\text{direction pointing to } \boldsymbol{z_t}} + \underbrace{\sigma_t\epsilon_t}_{\text{random noise}},\end{aligned} \tag{2}$$

where $\epsilon_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ is a standard Gaussian noise, α_t, σ_t specify a differentiable noise schedule at time step t and ϵ_θ is the U-Net with learnable parameters θ. Since DDIM is a deterministic sampling method, its denoising process is inherently reversible. DDIM inversion can be employed to map a clean latent $\boldsymbol{z_0}$ back to a noised latent $\boldsymbol{z_T}$:

$$\begin{aligned}\hat{\boldsymbol{z}}_{\boldsymbol{t}} &= \sqrt{\alpha_t}\left(\frac{\hat{\boldsymbol{z}}_{\boldsymbol{t-1}} - \sqrt{1-\alpha_{t-1}}\epsilon_\theta\left(\hat{\boldsymbol{z}}_{\boldsymbol{t-1}}, t-1\right)}{\sqrt{\alpha_{t-1}}}\right) + \\ &\sqrt{1-\alpha_t-\sigma_{t-1}^2}\cdot\epsilon_\theta\left(\hat{\boldsymbol{z}}_{\boldsymbol{t-1}}, t-1\right) + \sigma_{t-1}\epsilon_{t-1}.\end{aligned} \tag{3}$$

The inverted noisy latent $\hat{\boldsymbol{z}}_{\boldsymbol{t}}$ can be reconstructed into a clean latent $\hat{\boldsymbol{z}}_{\boldsymbol{0}}$ that closely resembles the original latent $\boldsymbol{z_0}$ through DDIM denoising.

ControlNet. ControlNet [6] is a versatile extension of the SD model, enabling control over the geometric, content, and structural attributes of generated images. It directly leverages trainable copies of the U-Net's down&middle blocks to extract features from conditioning maps. By introducing additional "zero convolution" layers and connecting them to the SD model, it effectively eliminates harmful noise during training while also stabilizing the training process. With the rapid growth of the diffusion model community in recent years, ControlNet now supports a wide range of conditioning types, including depth maps, segmentation maps, skeleton maps, canny maps, and scribbles. Additionally, the composition of multiple ControlNet modules is straightforward, involving the summation of their respective outputs before they are fed into the SD model.

Condition-specific LoRAs. To enable a single ControlNet to train on multiple conditions, CtrLoRA [7] introduces condition-specific LoRAs. Specifically, this structure integrates LoRA layers into each linear layer of ControlNet, as shown in Figure 1, and during the training of different tasks, the condition-specific LoRAs are switched to the corresponding LoRA instance. This allows different condition-specific LoRAs to extract task-specific features, while ControlNet captures the shared features across all tasks. Each LoRA layer consists of two linear layers (down and up layers), where the down layer reduces the dimensionality of the input, and the up layer subsequently restores its dimensions. For all LoRA layers, the down projection weights are initialized with a normal distribution scaled by $1/\text{rank}$, while the up projection weights are initialized to zero.

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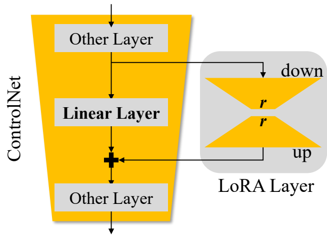

Fig. 1. The architecture of condition-specific LoRAs.

B. RELATED WORK

B.1. Human Image Animation

Given a reference image and a pose sequence (or driving video), human image animation aims to animate the reference image according to the specified pose sequence. With the strong generative capabilities of diffusion models, recent research has increasingly explored their use in HIA. Disco [8] tackles the problem by decoupling the control of foreground subjects and background. AnimateAnyone [9] introduces ReferenceNet to merge detailed features of the reference image into the denoising model. Methods such as MagicPose [10], MagicAnimate [11], Champ [12], and RealisDance [13] employ similar structures to [9]. However, Champ differs by using the SMPL model [14] to render depth maps, normal maps, and semantic maps as additional conditioning, while RealisDance utilizes SMPL colorful surface and HaMeR [15] as extra conditioning signals. MimicMotion [16] and StableAnimator [17] bypass ReferenceNet and directly add features extracted from the reference image via a VAE encoder [4] to the noisy latent. Most methods mentioned above replace the CLIP text encoder [18] in SD [3] or SVD [19] with an image encoder [18, 20], using cross-attention to integrate the reference image's global features into the denoising process. However, most existing approaches suffer from high computational cost and limited adaptability. Moreover, replacing the CLIP text encoder with image encoders introduces feature alignment issues unless the U-Net is retrained, which is resource-intensive. In contrast, Meta-Animator avoids architectural modifications by using text-based reference identifiers and condition-specific LoRAs. Guided by curated Meta-Tasks, this framework achieves efficient few-shot adaptation, strong identity preservation, and highly competitive performance.

B.2. Meta-Learning

Meta-learning [21], or "learning to learn," focuses on acquiring general prior knowledge from a variety of tasks, facilitating rapid adaptation to unseen data or tasks with only a few examples. This property makes it well-suited for few-shot learning scenarios [22, 23, 24]. Recently, there has been growing interest in combining meta-learning with diffusion models. MetaDiff [25] integrates meta-learning with conditional diffusion, modeling optimization as a denoising process to enhance task adaptation. Similarly, ProtoDiff [26] employs task-guided diffusion to generate prototypes for few-shot classification, improving generalization. Meta-ControlNet [27] applies meta-learning to facilitate rapid adaptation across tasks in diffusion models. In addition, recent work such as MetaDiffusion [28] explores unconditional diffusion models for meta-learning without explicit data, further expanding the applicability of meta-diffusion frameworks in low-data regimes. These studies collectively demonstrate that integrating meta-learning with diffusion architectures significantly improves task adaptation and performance in few-shot and generative tasks.

B.3. Subject-driven Generation.

Subject-driven generation aims to personalize image or video synthesis conditioned on limited subject-specific data. In image-based scenarios, DreamBooth [29] introduces a subject-specific token through fine-tuning to enable personalized text-to-image generation. Subsequent works enhance this paradigm by exploring optimized embedding strategies and disentangled representations. For instance, CatVersion [30] utilizes concatenation-based embedding fusion to improve identity fidelity, while [31] introduces textual regularization to mitigate overfitting and language drift. Extending these concepts to the temporal domain, DreamVideo [32] decouples the learning of subject appearance and motion into a two-stage pipeline. Similarly, VideoDreamer [33] addresses the challenging multi-subject scenario by employing Disen-Mix fine-tuning and motion disentanglement to faithfully render multiple subjects. These advancements demonstrate the feasibility of subject-driven generation and directly inspire the design of Meta-Animator, which leverages text-guided reference identifiers to achieve identity-preserving animation with minimal data.

C. IMPLEMENTATION DETAILS

All training was conducted using only the TikTok dataset [34], without any additional data. For a fair comparison, we evaluated our model on the TikTok test set as defined by DisCo [8]. We used DWPose [35] for skeleton maps, Grounded-SAM [36] for foreground segmentation maps, ArcFace [37] for face embeddings, and Sapiens [38] for body part segmentation, depth, and normal maps. All images were center-cropped to 512×512, the rank was set to 128 for all LoRAs, and the drop rate was set to 0.01. For consistency, we used SDv1.5 in all experiments. By default, a reference set includes 5 images, the scaling factor λ in DSH Loss is set to 0.1 [39]. We trained the image pre-training stage for 50k steps on 2 L20 GPUs with a learning rate of 1e-5 and batch size of 16. The motion module was trained for 10k steps in the temporal learning stage with the same settings. For few-shot fine-tuning, we trained the model for 500 steps on a single L20 GPU with a learning rate of 1e-5 and batch size of 8. During inference, we used the DDIM inversion of the first reference image as the initial noise for each frame, and performed 50 denoising steps with a DDIM sampler. For long video generation, we set the overlap between video segments to 4 frames, following the strategy proposed in MagicAnimate [11]. Furthermore, xFormers [40] are employed to optimize memory efficiency.

D. MORE DETAILS FOR META-TASKS

In Meta-Animator, we use a default set of 9 tasks, as shown in Figure 7, to form the Meta-Tasks. These include Depth, Normal, Fg-Inpainting, Segmentation, DWPose, and HaMeR Pose as foreground learning tasks, Bg-Outpainting as a background learning task, and Canny and Blur as global learning tasks. Each task corresponds to a condition-specific LoRA. Since LoRAs in the Meta-Animator architecture can be added or removed freely, the number and type of Meta-Tasks can be adjusted as needed. For example, to achieve higher animation quality, additional tasks such as HED, Lineart, or DensePose [41] can be incorporated. Conversely, to speed up training, the number of tasks can be reduced. However, it is important to note that the relationship between animation quality and the number

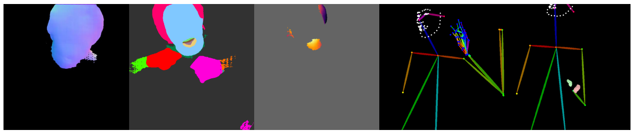

Fig. 2. Examples of low-quality or noisy conditioning maps generated by existing methods. From left to right: normal map, body-part segmentation map, and depth map (all generated using Sapiens); skeleton map (generated using DWPose); and hand pose map (from the HaMeR model). These maps exhibit typical failure cases, such as missing body parts, inaccurate segmentation boundaries, skeletal distortions, and incorrect estimation of hand size and pose, caused by algorithmic limitations and the low quality of video frames.

of tasks is not linear, and experiments are required to determine the impact of different tasks on animation quality.

E. DATA CURATION PROCESS

Following the protocol adopted in previous works, we use the first 335 video clips from TikTok dataset for training. Clips containing multiple foreground subjects are excluded to maintain training stability. Various conditioning maps are generated using DWPose [35], Grounded-SAM [36], Sapiens [38], and traditional image processing algorithms. However, due to limitations in these algorithms and the image quality of video frames, many of the generated conditioning maps are noisy or low quality, as illustrated in Figure 2. To avoid adverse effects on model training caused by these artifacts, we manually filter out low-quality conditioning maps and retain only those with satisfactory visual fidelity. In the Meta-Animator framework, reference identifiers serve as keys that associate reference images with their visual features in the model parameters. Therefore, it is crucial to maintain a one-to-one mapping between each reference identifier and its associated reference image to ensure stable training. Since some different video clips in the TikTok dataset belong to the same identity, we manually review the dataset to assign consistent reference identifiers to each unique identity. For few-shot fine-tuning, we introduce 20 additional reference sets alongside the target reference set during training. This strategy mitigates overfitting and reduces the impact of low-quality target references.

Reference Set Selection. The choice of reference images has a significant impact on the quality of the generated details. Selecting highly similar images, such as temporally adjacent frames, often leads to noticeable degradation in overall quality. To ensure fairness and diversity, we uniformly sample K reference images to construct the reference set during fine-tuning. Additionally, due to the prevalence of motion blur in the TikTok dataset, we replace severely blurred frames with the nearest temporally adjacent frames.

Reference Identifier. The model is prompted with the template: *"A person looks like [V]"*. Instead of adopting the DreamBooth [29] strategy of selecting rare tokens from the vocabulary and inverting them into text space, we opt for a simpler approach by constructing reference identifiers using common letters, numbers, and symbols. To mitigate tokenization issues, as addressed in DreamBooth, shorter identifiers are employed. Specifically, reference identifiers are formed using combinations of uppercase letters $(A - Z)$, lowercase letters $(a - z)$, digits $(1 - 9)$, and the special symbol "@". Examples of such identifiers include *A@8*, *q@7*, and *W@6*. To ensure a clear separation between training and evaluation data, the characters *T*, *t*, *U*, and *u* are excluded during training. Identifiers beginning with *T* or *t* are reserved for the TikTok test set, while those starting with *U* or *u* are designated for out-of-distribution test set.

F. QUANTITATIVE ANALYSIS ON TRAINING COST

Table 1 presents a detailed comparison of the training resources required by various animation methods. Compared to existing approaches, Meta-Animator achieves comparable animation quality while significantly reducing the computational burden and data dependency. Most competing methods rely on large-scale GPU clusters, such as 8 to 32 A100 or V100 GPUs, and are trained for tens of thousands to millions of steps. Several methods also incorporate extensive private datasets, including thousands of video clips. In contrast, Meta-Animator completes training using only two L20 GPUs with 48 GB of memory for training. Furthermore, it relies solely on the publicly available TikTok dataset, using 335 videos for pre-training and only 105 images during fine-tuning. These results demonstrate that Meta-Animator offers a more efficient and accessible solution for video animation, particularly in settings with limited computational resources.

G. ADDITIONAL EXPERIMENTS

G.1. Meta-Tasks during Few-shot Fine-tuning

We further isolate the effect of Meta-Tasks during few-shot fine-tuning while retaining the default task configuration during pre-training. As reported in Table 2, removing Segmentation, Depth, and Normal substantially reduces image quality, confirming the importance of diverse FLTs for reference-specific adaptation. Excluding BLTs or GLTs mainly increases FVD, indicating their contribution to temporal and structural consistency. Removing HaMeR Pose and Foreground Inpainting produces only minor changes, suggesting that their effects partially overlap with those of the remaining foreground tasks.

G.2. Text-Guided Property Modification

Since we use text-based reference identifiers to guide the model in generating specific foreground and background, and the denoising U-Net remains unchanged during training, the model should retain text-guided generation capabilities. This allows us to modify certain properties of the generated image by adjusting the text prompt. For instance, we alter the clothing color and hair color of the foreground identity. We prompt the model with prompts such as: *"A person looks like [V] and is wearing a black dress"* or *"A person looks like [V] and has blue hair"*. As shown in the first row of Figure 3, the generated images reflect the specified modifications while preserving consistency with the reference image.

However, the model does not always modify the reference image according to the text prompt in every generated sample. We find that the initial noise significantly affects the success rate of property modification, and the model struggles to make large-scale modifications to the reference image. For example, when provided with prompts such as *"A person looks like [V] and is wearing a black T-shirt"* or *"A person looks like [V] and is standing in front of the Eiffel Tower"*, the generated images, as shown in the second row of Fig. 3, fail to reflect the specified changes. We attribute this to the model being trained on a large number of [*reference identifier, reference image*] pairs, leading it to prioritize faithfully reconstructing the foreground and background of the reference image when encountering the reference identifier *[V]*, while overlooking certain modifications specified in the text prompt.

Table 1. Comparison of Training Resources Across Different Methods

Method	Training Stage	Training Resources	Dataset Size
DisCo [8]	Image Pre-training Video Fine-tuning	32 V100 GPUs for 25K steps 8 V100 GPUs for 70K steps	700K images TikTok 335 videos + 600 in-house videos
AnimateAnyone [9]	Image Pre-training Temporal Learning	4 A100 GPUs for 30K steps 4 A100 GPUs for 10K steps	TikTok 335 videos + 5K in-house videos TikTok 335 videos + 5K in-house videos
MagicPose [10]	Image Pre-training Image Fine-tuning Temporal Learning	8 A100 GPUs for 10K steps 8 A100 GPUs for 20K steps 8 A100 GPUs for 30K steps	TikTok 335 videos TikTok 335 videos TikTok 335 videos
Champ [12]	Image Pretraining Temporal Learning	8 A100 GPUs for 60K steps 8 A100 GPUs for 20K steps	TikTok 335 videos + 5K in-house videos TikTok 335 videos + 5K in-house videos
MimicMotion [16]	-	8 A100 GPUs for 6M steps	4,436 in-house videos (avg. 20.1s)
StableAnimator [17]	-	4 A100 80G GPUs for 33M steps	3,000 in-house videos (60∼90s)
Meta-Animator (Ours)	Image Pre-training Temporal Learning Few-shot Fine-tuning	2 L20 GPUs (48G) for 50K steps 2 L20 GPUs (48G) for 10K steps 1 L20 GPU (48G) for 500 steps	TikTok 335 videos TikTok 335 videos 21 reference sets (105 images)

Table 2. The ablation study in the Few-shot Fine-tuning stage evaluates the impact of Meta-Tasks on performance. *H.P.* refers to the HaMeR Pose task.

Exp.	FLTs (Foreground Learning Tasks)						BLTs	GLTs		Metrics			
	DWPose	H.P.	Fg-Inpaint	Seg.	Depth	Normal	Bg-Outpaint	Canny	Blur	FID↓	SSIM↑	PSNR↑	FVD↓
1 (All Tasks)	✓	✓	✓	✓	✓	✓	✓	✓	✓	**27.61**	**0.782**	**30.52**	**169.18**
2 (*w/o.* GLTs)	✓	✓	✓	✓	✓	✓	✓			29.82	0.761	29.64	195.25
3 (*w/o.* BLTs)	✓	✓	✓	✓	✓	✓		✓	✓	29.16	0.789	30.36	185.49
4 (*w/o.* BLTs&GLTs)	✓	✓	✓	✓	✓	✓				31.35	0.755	29.66	200.4
5 (*w/o.* FLTs except DWPose)	✓						✓	✓	✓	35.20	0.721	28.77	190.29
6 (*w/o.* Seg.&Depth&Normal)	✓	✓	✓				✓	✓	✓	33.94	0.738	28.98	180.18
7 (*w/o.* H.P.&Fg-Inpaint)	✓			✓	✓	✓	✓	✓	✓	27.81	0.784	30.45	164.60
8 (*w/ only* DWPose)	✓									41.34	0.715	28.91	219.62

Fig. 3. Property Modification Capability of Meta-Animator. The top row shows successful modifications of attributes like clothing and hair color while preserving identity and background. The bottom row shows failure cases where larger changes, such as altering clothing type or background, are not reflected. This is likely due to the model prioritizing reconstruction of the reference image when conditioned on "*[V]*".

G.3. Effect of Blur Operators

The blur image guided animation task uses blurred target images to provide global appearance and layout cues. We examine how different blur operators affect the quality of these conditioning images and the resulting model performance. As shown in Figure 4, average and median filtering produce relatively uniform blur patterns that suppress local details while preserving the overall scene structure. Bilateral filtering retains local edges selectively and introduces spatially uneven blur patterns, which create ambiguous conditioning signals during training. Models trained with these signals exhibit reduced visual fidelity and abnormal textures in the generated results.

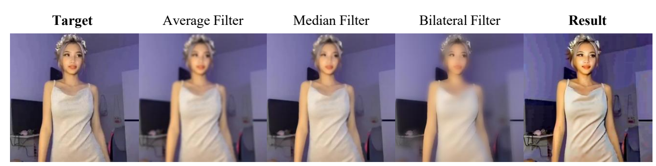

Fig. 4. Visualizations of images processed with different blur algorithms and the model's generated results using bilaterally filtered images during training. The average and median filters produce uniform and less structured blurring effects, while the bilateral filter introduces uneven blur artifacts. These inconsistencies can confuse the model during training, leading to degraded generation quality. The generated results show that bilateral-filtered inputs may lead to degraded fidelity and the emergence of abnormal textures.

G.4. Qualitative Analysis of DSH Loss.

Figure 5 further illustrates the effect of DSH Loss on visual quality and color consistency. Without DSH Loss, some generated samples exhibit abnormal color tones and blurred object boundaries, as shown in the second column. Applying DSH Loss produces colors that are more consistent with the target image and improves the clarity of local structures. The refrigerator and television boundaries in the third column provide representative examples of these improvements. These observations complement the main-paper ablation and demonstrate the benefit of joint latent-space and image-space supervision for preserving fine-grained visual details.

Fig. 5. Ablation study on DSH Loss. Without DSH Loss, the outputs show color shifts and blurry background edges. With DSH Loss, the model produces more accurate color tones and clearer boundaries, especially around facial regions and background objects.

G.5. Reference-Set Comparisons

DreamBooth [29] suggests that for common subjects that lie strongly in the diffusion model's distribution, such as a Corgi dog, high-fidelity appearance can be captured with as few as one or two images, given appropriate tuning. However, the objective of Meta-Animator and DreamBooth differs. While DreamBooth aims to recontextualize a subject based on a given text prompt, Meta-Animator focuses on animating the subject according to a given pose sequence. This requires the model not only to distinguish between the foreground and background in the reference image but also to establish an implicit mapping between the subject's motion and the pose sequence using the reference set. Consequently, we hypothesize that providing the model with more reference images for fine-tuning enhances animation quality. As shown in Table 3, we find that providing more reference images for fine-tuning improves animation quality. Considering the trade-off between practicality and performance, Meta-Animator adopts 5 reference images as the default reference set for fine-tuning. As illustrated in Figure 6, using a larger reference set as prior information enables more precise detail preservation, compared to methods where inference is performed using only a single reference image.

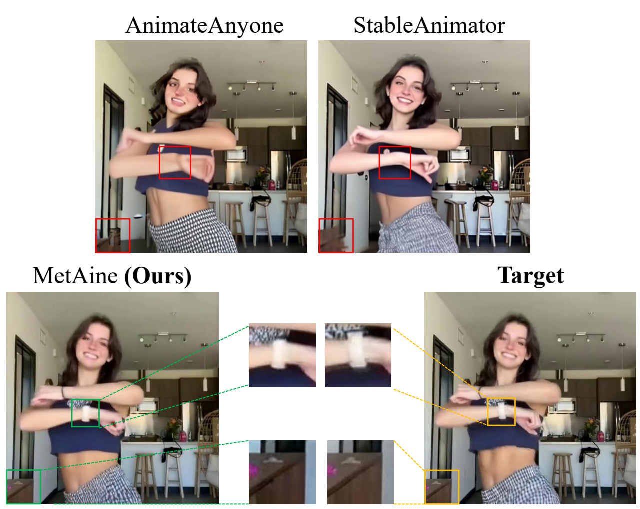

Fig. 6. Visualizations on detail preservation ability between Meta-Animator and baselines. Competing methods (top row) rely on a single reference image during inference, often leading to missing or distorted details, as shown in the red boxes. In contrast, Meta-Animator (bottom left), fine-tuned with reference set, retains fine-grained appearance details and structural consistency, as highlighted in the green boxes.

Table 3. Ablation study of Reference Set.

Reference Image Num.	FID↓	SSIM↑	PSNR↑	FVD↓
1	32.25	0.721	29.17	183.14
20	26.92	0.790	30.69	165.76
All (100 $\sim$ 600)	**23.04**	**0.814**	**31.94**	**156.50**
Meta-Animator (5)	27.61	0.782	30.52	169.18

H. DISCUSSION

Meta-Animator achieves competitive performance in human image animation with significantly fewer training resources. Although it falls slightly short of several state-of-the-art approaches in temporal consistency metrics such as FVD, its efficiency offers clear advantages in practical applications. Furthermore, the design of Meta-Animator makes it highly suitable for integration with more powerful video generation frameworks, including Wan2.1 [42] and CogVideoX [43]. These models have massive parameter scale (e.g., Wan2.1: 14B, CogVideoX: 5B), making it impractical to create a separate trainable copy (ReferenceNet) to extract reference image features. In contrast, Meta-Animator requires no architectural changes to the denoising model or additional ReferenceNet. Its use of text-based reference identifiers and LoRA fine-tuning makes it easily adaptable to larger video generation models. It is also worth noting that our method supports fine-tuning with multiple reference images. Simply increasing the number of reference images during the third stage (Few-shot Fine-tuning stage) leads to noticeable improvements in generation quality. This is a highly practical advantage in real-world applications. In contrast, most HIA methods support only a single reference image during inference, which inherently limits their applicability in certain scenarios.

I. MORE VISUALIZATION RESULTS

Figure 8 compares Meta-Animator with representative baselines on the TikTok dataset. Meta-Animator achieves identity preservation and pose transfer comparable to StableAnimator. DSH Loss helps retain fine-grained facial and appearance details. The second row also shows occasional motion blur inherited from the TikTok training data, suggesting that pretraining on higher-quality human images could further improve sharpness. Figures 9, 10, and 11 provide additional results on the TikTok and DanceIt datasets.

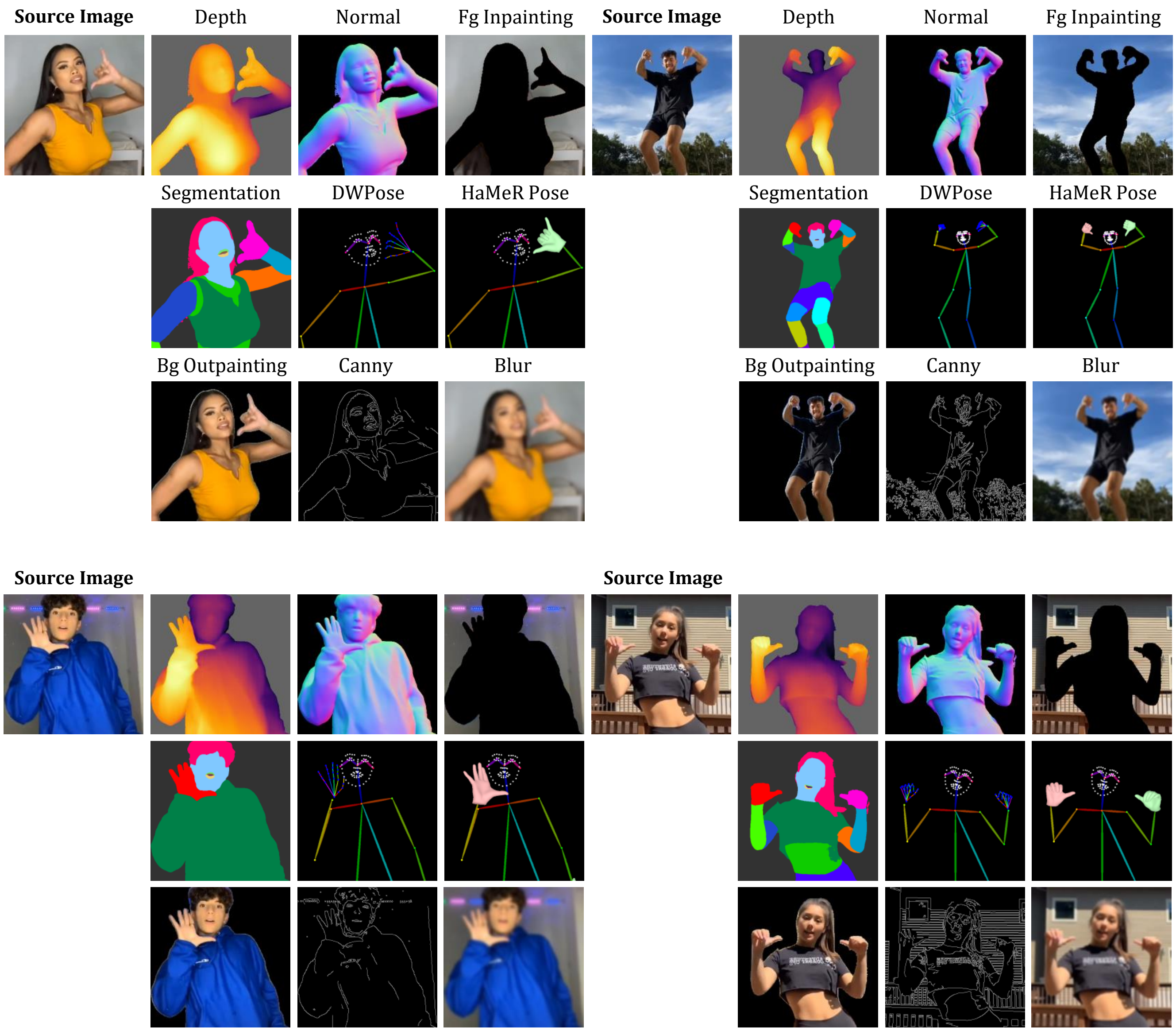

Fig. 7. More example data from Meta-Tasks.

Fig. 8. Qualitative comparisons between Meta-Animator and baselines on the TikTok dataset. The conditioning pose map is overlaid in the top corner.

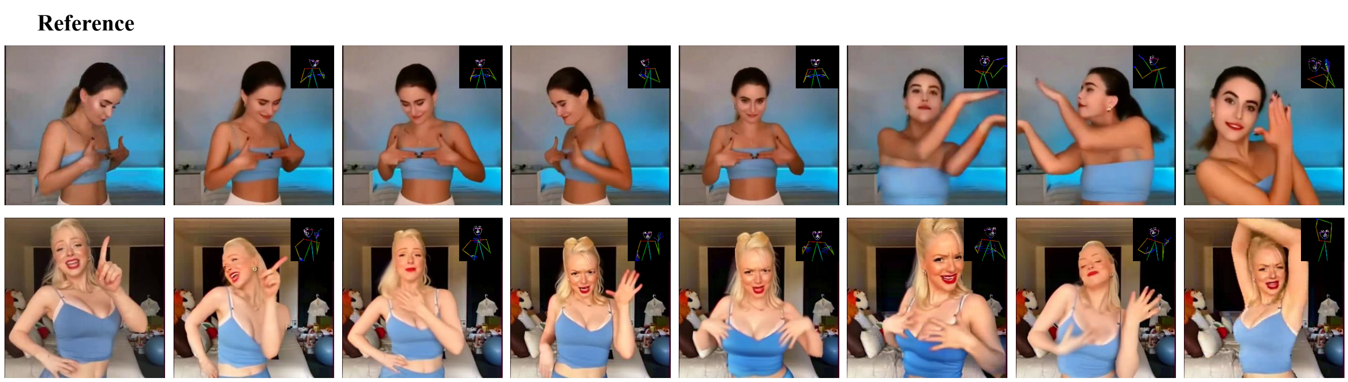

Fig. 9. Sample results of Meta-Animator on TikTok dataset. The conditioning pose map is overlaid in the top corner. Our model demonstrates good temporal consistency across frames while preserving both the subject's identity and the background. Even in challenging cases involving fine hand gestures or large body movements, Meta-Animator produces stable and coherent animations.

Fig. 10. Sample results of Meta-Animator on TikTok test set.

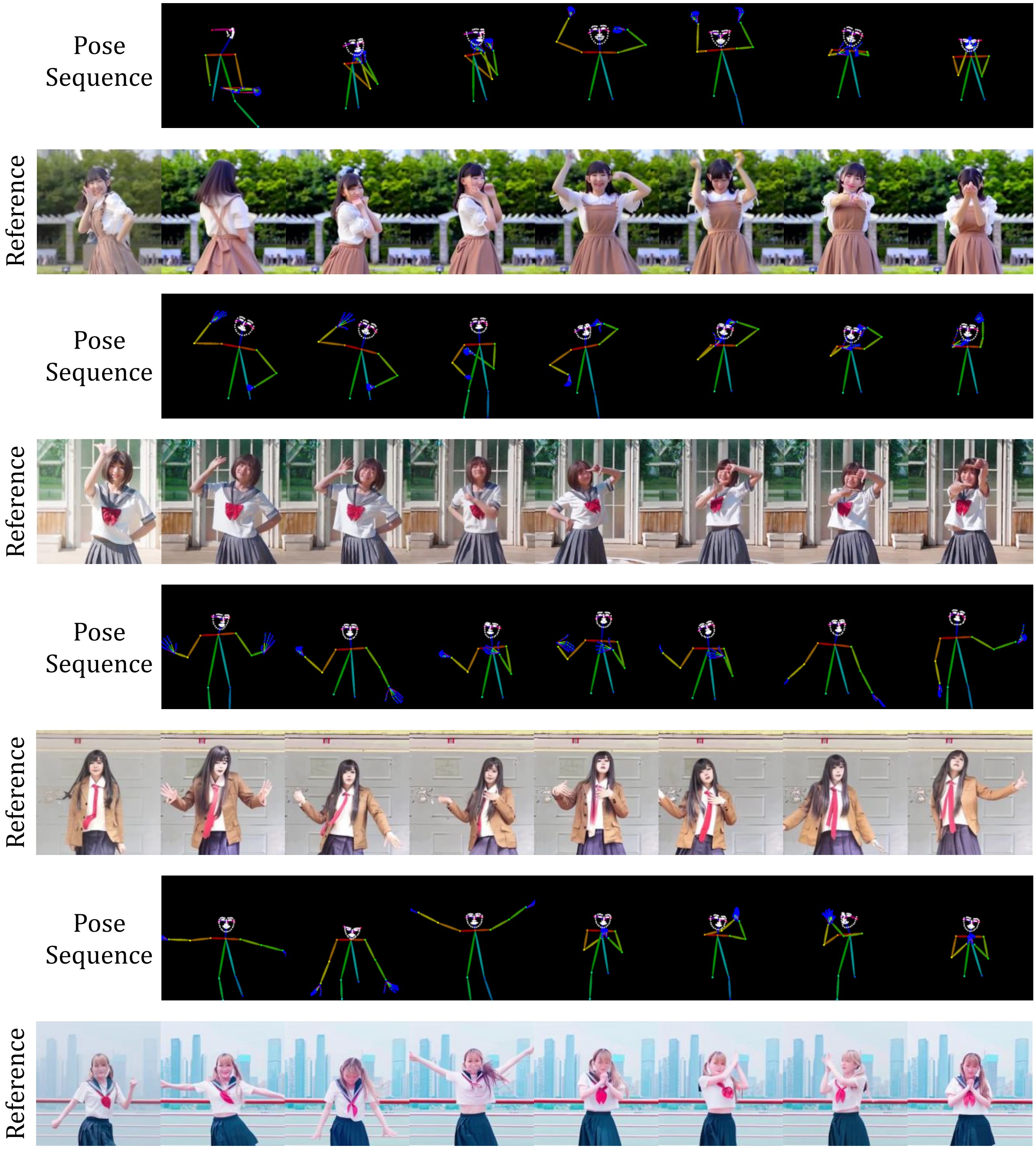

Fig. 11. Sample results of Meta-Animator on DanceIt dataset.

J. REFERENCES

[1] Jonathan Ho, Ajay Jain, and Pieter Abbeel, "Denoising diffusion probabilistic models," *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.

[2] Jiaming Song, Chenlin Meng, and Stefano Ermon, "Denoising diffusion implicit models," *arXiv preprint arXiv:2010.02502*, 2020.

[3] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10684–10695.

[4] Aaron Van Den Oord, Oriol Vinyals, et al., "Neural discrete representation learning," *Advances in neural information processing systems*, vol. 30, 2017.

[5] Olaf Ronneberger, Philipp Fischer, and Thomas Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18*. Springer, 2015, pp. 234–241.

[6] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 3836–3847.

[7] Yifeng Xu, Zhenliang He, Shiguang Shan, and Xilin Chen, "Ctrlora: An extensible and efficient framework for controllable image generation," *arXiv preprint arXiv:2410.09400*, 2024.

[8] Tan Wang, Linjie Li, Kevin Lin, Yuanhao Zhai, Chung-Ching Lin, Zhengyuan Yang, Hanwang Zhang, Zicheng Liu, and Lijuan Wang, "Disco: Disentangled control for realistic human dance generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 9326–9336.

[9] Li Hu, "Animate anyone: Consistent and controllable image-to-video synthesis for character animation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 8153–8163.

[10] Di Chang, Yichun Shi, Quankai Gao, Hongyi Xu, Jessica Fu, Guoxian Song, Qing Yan, Yizhe Zhu, Xiao Yang, and Mohammad Soleymani, "Magicpose: Realistic human poses and facial expressions retargeting with identity-aware diffusion," in *Forty-first International Conference on Machine Learning*, 2023.

[11] Zhongcong Xu, Jianfeng Zhang, Jun Hao Liew, Hanshu Yan, Jia-Wei Liu, Chenxu Zhang, Jiashi Feng, and Mike Zheng Shou, "Magicanimate: Temporally consistent human image animation using diffusion model," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 1481–1490.

[12] Shenhao Zhu, Junming Leo Chen, Zuozhuo Dai, Zilong Dong, Yinghui Xu, Xun Cao, Yao Yao, Hao Zhu, and Siyu Zhu, "Champ: Controllable and consistent human image animation with 3d parametric guidance," in *European Conference on Computer Vision*. Springer, 2024, pp. 145–162.

[13] Jingkai Zhou, Benzhi Wang, Weihua Chen, Jingqi Bai, Dongyang Li, Aixi Zhang, Hao Xu, Mingyang Yang, and Fan Wang, "Realisdance: Equip controllable character animation with realistic hands," *arXiv preprint arXiv:2409.06202*, 2024.

[14] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J Black, "Smpl: A skinned multi-person linear model," in *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, pp. 851–866. 2023.

[15] Georgios Pavlakos, Dandan Shan, Ilija Radosavovic, Angjoo Kanazawa, David Fouhey, and Jitendra Malik, "Reconstructing hands in 3d with transformers," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 9826–9836.

[16] Yuang Zhang, Jiaxi Gu, Li-Wen Wang, Han Wang, Junqi Cheng, Yuefeng Zhu, and Fangyuan Zou, "Mimicmotion: High-quality human motion video generation with confidence-aware pose guidance," *arXiv preprint arXiv:2406.19680*, 2024.

[17] Shuyuan Tu, Zhen Xing, Xintong Han, Zhi-Qi Cheng, Qi Dai, Chong Luo, and Zuxuan Wu, "Stableanimator: High-quality identity-preserving human image animation," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 21096–21106.

[18] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al., "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PMLR, 2021, pp. 8748–8763.

[19] Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al., "Stable video diffusion: Scaling latent video diffusion models to large datasets," *arXiv preprint arXiv:2311.15127*, 2023.

[20] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al., "Dinov2: Learning robust visual features without supervision," *arXiv preprint arXiv:2304.07193*, 2023.

[21] Chelsea Finn, Pieter Abbeel, and Sergey Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *International conference on machine learning*. PMLR, 2017, pp. 1126–1135.

[22] Oriol Vinyals, Charles Blundell, Timothy Lillicrap, Daan Wierstra, et al., "Matching networks for one shot learning," *Advances in neural information processing systems*, vol. 29, 2016.

[23] Jake Snell, Kevin Swersky, and Richard Zemel, "Prototypical networks for few-shot learning," *Advances in neural information processing systems*, vol. 30, 2017.

[24] Eleni Triantafillou, Tyler Zhu, Vincent Dumoulin, Pascal Lamblin, Utku Evci, Kelvin Xu, Ross Goroshin, Carles Gelada, Kevin Swersky, Pierre-Antoine Manzagol, et al., "Meta-dataset: A dataset of datasets for learning to learn from few examples," *arXiv preprint arXiv:1903.03096*, 2019.

[25] Baoquan Zhang, Chuyao Luo, Demin Yu, Xutao Li, Huiwei Lin, Yunming Ye, and Bowen Zhang, "Metadiff: Meta-learning with conditional diffusion for few-shot learning," in *Proceedings of the AAAI conference on artificial intelligence*, 2024, vol. 38, pp. 16687–16695.

[26] Yingjun Du, Zehao Xiao, Shengcai Liao, and Cees Snoek, “Protodiff: Learning to learn prototypical networks by task-guided diffusion,” *arXiv preprint arXiv:2306.14770*, 2023.

[27] Junjie Yang, Jinze Zhao, Peihao Wang, Zhangyang Wang, and Yingbin Liang, “Meta controlnet: Enhancing task adaptation via meta learning,” *arXiv preprint arXiv:2312.01255*, 2023.

[28] Yongxian Wei, Zixuan Hu, Li Shen, Zhenyi Wang, Lei Li, Yu Li, and Chun Yuan, “Meta-learning without data via unconditional diffusion models,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 11, pp. 11874–11885, 2024.

[29] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman, “Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 22500–22510.

[30] Ruoyu Zhao, Mingrui Zhu, Shiyin Dong, De Cheng, Nannan Wang, and Xinbo Gao, “Catversion: Concatenating embeddings for diffusion-based text-to-image personalization,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 6, pp. 6047–6058, 2025.

[31] Zhicai Wang, Ouxiang Li, Tan Wang, Longhui Wei, Yanbin Hao, Xiang Wang, and Qi Tian, “Prior preserved text-to-image personalization without image regularization,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 2, pp. 1318–1330, 2024.

[32] Yujie Wei, Shiwei Zhang, Zhiwu Qing, Hangjie Yuan, Zhiheng Liu, Yu Liu, Yingya Zhang, Jingren Zhou, and Hongming Shan, “Dreamvideo: Composing your dream videos with customized subject and motion,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 6537–6549.

[33] Hong Chen, Xin Wang, Guanning Zeng, Yipeng Zhang, Yuwei Zhou, Feilin Han, Yaofei Wu, and Wenwu Zhu, “Videodreamer: Customized multi-subject text-to-video generation with disen-mix finetuning on language-video foundation models,” *IEEE Transactions on Multimedia*, 2025.

[34] Yasamin Jafarian and Hyun Soo Park, “Learning high fidelity depths of dressed humans by watching social media dance videos,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 12753–12762.

[35] Zhendong Yang, Ailing Zeng, Chun Yuan, and Yu Li, “Effective whole-body pose estimation with two-stages distillation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4210–4220.

[36] Tianhe Ren, Shilong Liu, Ailing Zeng, Jing Lin, Kunchang Li, He Cao, Jiayu Chen, Xinyu Huang, Yukang Chen, Feng Yan, et al., “Grounded sam: Assembling open-world models for diverse visual tasks,” *arXiv preprint arXiv:2401.14159*, 2024.

[37] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou, “Arcface: Additive angular margin loss for deep face recognition,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 4690–4699.

[38] Rawal Khirodkar, Timur Bagautdinov, Julieta Martinez, Su Zhaoen, Austin James, Peter Selednik, Stuart Anderson, and Shunsuke Saito, “Sapiens: Foundation for human vision models,” in *European Conference on Computer Vision*. Springer, 2024, pp. 206–228.

[39] Rang Meng, Xingyu Zhang, Yuming Li, and Chenguang Ma, “Echomimicv2: Towards striking, simplified, and semi-body human animation,” *arXiv preprint arXiv:2411.10061*, 2024.

[40] Benjamin Lefaudeux, Francisco Massa, Diana Liskovich, Wenhan Xiong, Vittorio Caggiano, Sean Naren, Min Xu, Jieru Hu, Marta Tintore, Susan Zhang, Patrick Labatut, Daniel Haziza, Luca Wehrstedt, Jeremy Reizenstein, and Grigory Sizov, “xformers: A modular and hackable transformer modelling library,” `https://github.com/facebookresearch/xformers`, 2022.

[41] Rıza Alp Güler, Natalia Neverova, and Iasonas Kokkinos, “Densepose: Dense human pose estimation in the wild,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 7297–7306.

[42] Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, et al., “Wan: Open and advanced large-scale video generative models,” *arXiv preprint arXiv:2503.20314*, 2025.

[43] Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang, Wenyi Hong, Xiaohan Zhang, Guanyu Feng, et al., “Cogvideox: Text-to-video diffusion models with an expert transformer,” *arXiv preprint arXiv:2408.06072*, 2024.